Joint Low-Rank and Sparse Tensors Recovery for Video Synthetic Aperture Radar Imaging

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***Abstract*— Video synthetic aperture radar (SAR) receives more and more attention in recent years because it can provide continuous images of the observed scene. However, the enormous data of video SAR to obtain the multiframe images bring big challenges to its transmission, storage, and processing, especially for small unmanned aerial vehicle (UAV) platform. In this article, we aim at proposing an efficient video formation method for video SAR systems with reduced data. First, the characteristics of video SAR observed scene are analyzed. It is found that the observed scene with multiple frames can be modeled as the sum of a low-rank tensor and a sparse tensor efficiently. After that, the video formation problem for video SAR is modeled as a joint low-rank and sparse tensors recovery problem. Finally, an efficient tensor alternating direction method of multiplier is proposed to obtain the final SAR video. Compared with the traditional frequency- or time-domain imaging methods, the amount of data samples can be greatly reduced. On the other hand, the proposed method outperforms the state-of-the-art SAR imaging methods with reduced samples, including the joint low-rank and sparse matrices recovery method and the low-rank tensor recovery method. Numerical simulations validate the effectiveness of the proposed method.**

***Index Terms*— Imaging, low rank, sparse, synthetic aperture radar (SAR), tensor recovery, video SAR.**

I. Introduction

DIFFERENT from the optical sensors, synthetic aperture radar (SAR) illuminates the observed scene on its own first and receives the reflected signal from the observed scene, which enables it to have the capability of imaging under anytime of the day [1], [2]. Meanwhile, SAR system can penetrate the cloud and fog easily benefiting from its signal wavelength, which enables it to acquire images under any weather conditions [3]–[5]. Therefore, SAR has been widely applied in both military and civil applications. Generally, SAR system obtains one image once for the observed scene. Recently, a video SAR system is proposed to acquire the dynamic information of the observed area. Through observing the scene at certain video frame rate with a straight or circular flight, video SAR can form a sequence of SAR images for a scene of interest [6]–[8]. The video or image sequences can be applied to long-time surveillance and change detection of the ground area [9]–[12]. As a specific change detection application, ground moving target indication (GMTI) or ground moving target (GMT) tracking in video SAR has received increasing attention [8], [13], [14]. It has been found that GMTI can benefit a lot from the multiple frames obtained by video SAR [15]–[17].

Since the video of the observed scene is formed frame by frame, an intuitive way to achieve video SAR imaging is focusing each SAR frame separately. For each frame, the imaging processing is the same as the conventional SAR. Therefore, the efficient and effective frequency-domain imaging algorithms, such as range-Doppler algorithm and chirp scaling algorithm [18], can be applied. However, the frame interval of the SAR video is much smaller than the integration time to ensure a better azimuth resolution and sufficient dynamic information, which leads to the significant overlap of the aperture to form the consecutive frames. To make full use of the overlap aperture and reduce the redundant operations, the backprojection (BP) method was applied to video SAR imaging in [6]. To accelerate the processing, fast BP (FBP) algorithm utilizing subaperture division and recursive manner can be applied in [19]. Furthermore, parallel processing based on GPU can also be incorporated in the FBP algorithm for real-time operation applications.

However, the enormous amount of samples in video SAR still brings huge burden for the data storage, transmission, and processing, especially for the mini unmanned aerial vehicle (UAV) platform and small satellite platform, which seriously restricts the applications of video SAR systems. Compressed sensing (CS) technique states that a signal that is sparse (or sparse in some basis) can be recovered from highly incomplete samples or measurements than that required by the Shannon–Nyquist sampling theory [20]–[22]. Therefore, the CS technique to conventional SAR imaging was applied in [24]–[29] where the amount of samples can be greatly reduced. In addition to CS, matrix sensing (MS) theory was also applied to SAR imaging [29], [30]. MS utilizes the inertial correlation of matrix, i.e., low-rank information, to recovery the image [31], [32]. Compared with the CS method, MS is

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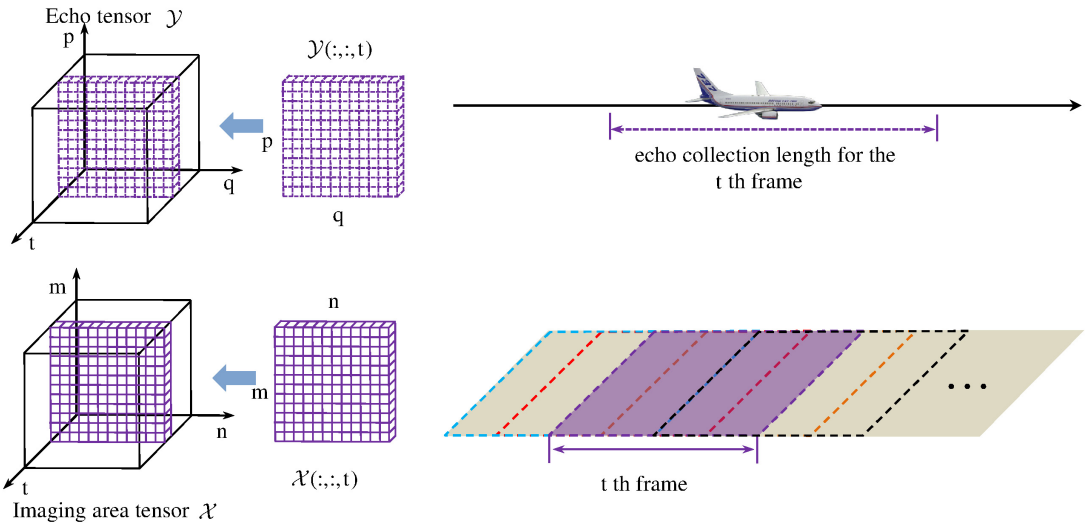


Fig. 1. Observation geometry and echo acquisition process of video SAR.

more effective because the inertial structure of the matrix is maintained. Furthermore, SAR imaging was modeled as a problem of joint sparse and low-rank matrices recovery based on the analysis of the sparsity and low rankness of the SAR image [33], [34]. However, all these methods are designed for conventional SAR imaging where each of frame data is processed independently. Therefore, the correlation between frames caused by the overlapping of observed scene is underutilized.

Recently developed tensor analysis and tensor recovery technique has attracted significant attention because it can fully exploit the internal structure of the high-dimensional data [35]–[37]. It has been widely applied to high-dimensional data processing, such as video analysis and color image restoration. The results show its superior performance compared to CS and MS methods. Essentially, the video to be formed in video SAR is a 3-D tensor, i.e., range, azimuth, and time. Therefore, the tensor analysis and recovery method can be applied to video SAR imaging. In [38], a low-rank tensor recovery method was first applied to video SAR imaging where the observed scene to be recovered is modeled as a low-rank tensor. However, this method suffers performance degradation when the imaging scene exists a small portion of anomaly dominant scatterers. In these cases, the low rankness of video data is broken down by the anomaly dominant scatterers.

In this article, a novel strategy exploiting joint sparsity and low rankness of the observed scene tensor for video SAR imaging is proposed. First, the sparsity and low rankness of the real SAR video are analyzed. It is found that the SAR video can be modeled as the sum of a sparse tensor and a low-rank tensor, which is much more efficient than the low-rank tensor model. After that, a video SAR imaging problem is transformed into a joint l_0-minimization and tensor nuclear norm minimization problem based on the inverse processing of efficient frequency-domain imaging methods. Finally, a tensor alternating direction method of multiplier (ADMM) is proposed. Compared with the traditional frequency- or time-domain imaging methods, the amount of samples can be greatly reduced. On the other hand, the proposed method outperforms the state-of-the-art video SAR imaging algorithms.

The rest of this article is organized as follows. Section II analyzes the property of the SAR video and formulates the imaging problem of video SAR as a joint sparse and low-rank tensors recovery problem. The imaging method based on tensor ADMM is proposed in Section III. Simulation results are provided in Section IV to verify the effectiveness of the proposed method. Conclusions are given in Section V.

The following mathematical notations are adopted in the description of the following algorithm. We denote tensors by boldface Euler script letters, e.g., $\mathcal{X}$. Matrices are denoted by boldface capital letters, e.g., $\mathbf{X}$. Vectors are denoted by boldface lowercase letters, e.g., $\mathbf{x}$, and scalars are denoted by lowercase letters, e.g., x. For a three-way tensor $\mathcal{X} \in \mathbb{C}^{M\times N\times T}$, its (m,n,t)th entry is denoted as $\mathcal{X}_{mnt}$ or x_{mnt}. $\mathcal{X}(m,:,:)$, $\mathcal{X}(:,n,:)$, and $\mathcal{X}(:,:,t)$ represent the horizontal, lateral, and frontal slice of the tensor, respectively, which are matrices. Also, the frontal slice $\mathcal{X}(:,:,t)$ is denoted compactly as $\mathbf{X}^{(t)}$. $\lfloor x \rfloor$ is denoted as the nearest integer less than or equal to x and $\lceil x \rceil$ represents the one greater or equal to x.

II. Observation Model and Problem Modeling

First, the observation geometry of video SAR is introduced and the observation model is established. After that, the property of the real SAR video is analyzed and the imaging problem of video SAR is modeled as a joint low-rank and sparse tensors recovery problem.

A. Video SAR Observation Geometry and Imaging Model

The observation geometry for video SAR imaging is shown in Fig. 1. The radar platform travels along its straight flight path. In the meantime, the radar transmits linear frequency-modulated (LFM) pulses at a constant rate and receives the echo reflected by the observed area.

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In video SAR imaging, there are multiple frames need to be formed to reveal the dynamic information of the observed area. In the echo acquisition of the tth frame where $t = 1, 2, \ldots, T$ indexes the SAR video frames and T is the number of frames, we assume that the imaging area of tth frame is rectangular, with M pixels in azimuth and N pixels in range. The imaging area of tth frame is denoted by $\mathcal{X}(:, :, t)$, which is an $M \times N$ matrix and whose (m, n) entry is $\mathcal{X}(m, n, t)$. Meanwhile, let the echo of tth frame be denoted as $\mathcal{Y}(:, :, t) \in \mathbb{C}^{P\times Q}$, where P and Q indicate the total sample number of the azimuth time and range time, respectively.

The received signal of the tth frame at the pth azimuth pulse and qth range sequence is denoted as $\mathcal{Y}(p, q, t)$. The relation between the echo $\mathcal{Y}(p, q, t)$ and reflection coefficients $\mathcal{X}(:, :, t)$ of the tth frame is given as follows:

$$\mathcal{Y}(p, q, t) = \sum_{m=1}^{M}\sum_{n=1}^{N} \mathcal{X}(m, n, t)\varphi(p, q, m, n, t) \tag{1}$$

where

$$\begin{aligned}\varphi(p, q, m, n, t) = {} & \omega_r\left(\tau_q - \frac{2R(p, m, n, t)}{c}\right)\omega_a\left(\frac{p}{T_a}\right) \\ & \times \exp\left\{-j4\pi\frac{R(p, m, n, t)}{\lambda}\right\} \\ & \times \exp\left\{j\pi K_r\left(\tau_q - \frac{2R(p, m, n, t)}{c}\right)^2\right\}.\end{aligned} \tag{2}$$

Note that $\omega_r(\cdot)$ and $\omega_a(\cdot)$ are the range and azimuth envelope functions, respectively. $R(p, m, n, t)$ is the instant range between the targets located at position (m, n) at azimuth pulse p for the tth frame. λ is the wavelength of the transmitted signal, c is the speed of the light, K_r is the range chirp rate, and τ_q is the range sequence at qth sampling.

However, it is time-consuming and inefficient to calculate $\varphi(p, q, m, n, t)$ since there are nearly $PQ \times MN$ elements that need to be calculated separately for tth frame.

For each frame in video SAR, the echo collection model is the same as the conventional SAR system. In order to establish the echo collection model in a more efficient way, the inverse process of the effective and efficient imaging algorithms [23], [24], [39], such as range-Doppler algorithm and chirp scaling algorithm, was applied.

The imaging process of chirp scaling algorithm, i.e., the way to establish the mapping from the echo $\mathbf{Y}^{(t)}$ to the tth frame image $\mathbf{X}^{(t)}$, can be written as follows:

$$\begin{aligned}\mathbf{X}^{(t)} &= \mathfrak{I}\left(\mathbf{Y}^{(t)}\right) \\ &= \mathfrak{F}_a^{-1}\left\{\mathfrak{P}_3^{(t)}\left\{\mathfrak{F}_r^{-1}\left\{\mathfrak{P}_2^{(t)}\left\{\mathfrak{F}_r\left\{\mathfrak{P}_1^{(t)}\left\{\mathfrak{F}_a\left\{\mathbf{Y}^{(t)}\right\}\right\}\right\}\right\}\right\}\right\}\right\}\end{aligned} \tag{3}$$

where $\mathfrak{F}_r$ and $\mathfrak{F}_a$ represent the range and azimuth Fourier transforms, respectively, $\mathfrak{F}_r^{-1}$ and $\mathfrak{F}_a^{-1}$ represent the range and azimuth inverse Fourier transforms, respectively, and $\mathfrak{P}_1^{(t)}$, $\mathfrak{P}_2^{(t)}$, and $\mathfrak{P}_3^{(t)}$ are the phase multiplication processing of chirp scaling, range focusing, and azimuth compression for the tth frame image, respectively.

In [24], (3) was rewritten as a matrix multiplication manner

$$\mathrm{vec}\left(\mathbf{X}^{(t)}\right) = \mathbf{M}\mathrm{vec}\left(\mathbf{Y}^{(t)}\right) \tag{4}$$

where

$$\mathbf{M} = \mathbf{F}_a^H\mathbf{P}_3\mathbf{F}_r^H\mathbf{P}_2\mathbf{F}_r\mathbf{P}_1\mathbf{F}_a. \tag{5}$$

Note that $\mathbf{F}_r$ and $\mathbf{F}_a$ represent the matrix representation of Fourier transform with respect to range and azimuth directions, respectively, $\mathbf{F}_r^H$ and $\mathbf{F}_a^H$ represent the matrix representation of inverse Fourier transform with respect to range and azimuth directions, respectively, and $\mathbf{P}_i$ is the matrix representation of the phase function $\mathfrak{P}_i$. The detailed definitions of the matrices can be found in [24].

After that, the echo collection model (observation model) can be formed based on the inverse of the matrix multiplication manner of imaging process. The matrix-based observation model can be given as

$$\mathrm{vec}\left(\mathbf{Y}^{(t)}\right) = \mathbf{M}^{-1}\mathrm{vec}\left(\mathbf{X}^{(t)}\right) \tag{6}$$

where

$$\mathbf{M}^{-1} = \mathbf{F}_a^H\mathbf{P}_1^H\mathbf{F}_r^H\mathbf{P}_2^H\mathbf{F}_r\mathbf{P}_3^H\mathbf{F}_a. \tag{7}$$

Compared with the point-by-point calculation of the relation between $\mathbf{X}^{(t)}$ and $\mathbf{Y}^{(t)}$ as shown in (1), the generation of the matrix-based observation model $\mathbf{M}^{-1}$ is computationally efficient by applying the inverse of the fast imaging algorithm. The CS- and MS-based imaging methods benefit a lot from the ingenious formation of the observation model. However, computing and storing $\mathbf{M}^{-1}$ require large amounts of computing resources and storage space because $\mathbf{M}^{-1} \in \mathbb{C}^{PQ\times MN}$, which extremely limits the practical applications of the matrix-based observation model.

Indeed, the purpose of forming observation model is to find the relationship between the tth frame image $\mathbf{X}^{(t)}$ and the echo $\mathbf{Y}^{(t)}$. Thus, the observation model can be directly written as the inverse of chirp scaling algorithm processing steps, which is given in (8), as shown at the bottom of the page. In (8), $(\cdot)^{-1}$ represents the inverse process and $(\cdot)^*$ is the conjugate operation.

Instead of using the computationally expensive matrix-based observation model to establish the mapping from imaging scene frame to its corresponding echo, the decoupling model $\mathfrak{R}(\cdot)$ is applied as the mapping from imaging scene to echo. $\mathfrak{R}(\cdot)$ is computationally efficient since the calculation and storage of the high-dimensional matrix are avoided. The dimension of the decoupling model is almost the same as that of the observed scene.

Applying the observation model in (8), the relation between the imaging scene $\mathbf{X}^{(t)}$ and echo $\mathbf{Y}^{(t)}$ of the tth frame is established. Furthermore, the relation between the imaging scene tensor $\mathcal{X}$ and the video SAR echo $\mathcal{Y}$ can be easily

$$\mathbf{Y}^{(t)} = \mathfrak{R}\left(\mathbf{X}^{(t)}\right) = \mathfrak{F}_a^{-1}\left\{\left(\mathfrak{P}_1^{(t)}\right)^*\left\{\mathfrak{F}_r^{-1}\left\{\left(\mathfrak{P}_2^{(t)}\right)^*\left\{\mathfrak{F}_r\left\{\left(\mathfrak{P}_3^{(t)}\right)^*\left\{\mathfrak{F}_a\left\{\mathbf{X}^{(t)}\right\}\right\}\right\}\right\}\right\}\right\}\right\} \tag{8}$$

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obtained by rearranging the multiple frames of images and echo matrices as tensors, respectively, which is given as

$$\mathcal{Y}(:,:,t) = \Re(\mathcal{X}(:,:,t)), \quad t = 1, 2, \ldots, T. \tag{9}$$

The subsampling of the tth frame echo in azimuth and range can be regarded as the operation that a reduced set of azimuth and range elements in echo matrix $\mathbf{Y}^{(t)}$ is taken. Note that the reduced set is randomly taken. $\mathbf{\Theta}_a$ and $\mathbf{\Theta}_r$ are denoted as the subsampling matrices along the azimuth and range directions, respectively. After subsampling, the reduced echo received by the receiver is given as

$$\mathcal{Y}_d(:,:,t) = \mathbf{\Theta}_a \mathcal{Y}(:,:,t)\mathbf{\Theta}_r = \mathbf{\Theta}_a \Re(\mathcal{X}(:,:,t))\mathbf{\Theta}_r \quad t = 1, 2, \ldots, T. \tag{10}$$

B. SAR Video Property

To recover the original SAR video from the reduced sampled echo, the CS theory applies the sparsity property of the data. The original SAR video $\text{vec}(\mathcal{X})$ can be recovered by minimizing the following l_0-norm problem, namely:

$$\begin{aligned} &\arg \min_{\text{vec}(\mathcal{X})} \ \|\text{vec}(\mathcal{X})\|_0 \\ &\text{s.t. } \mathcal{Y}_d(:,:,t) = \mathbf{\Theta}_a \Re(\mathcal{X}(:,:,t))\mathbf{\Theta}_r, \quad t = 1, 2, \ldots, T. \end{aligned} \tag{11}$$

The minimization problem in (11) is a nondeterministic (NP) hard problem. In the CS theory, the previous problem can be solved using greedy strategies, such as orthogonal matching pursuit (OMP) [40], or converting the l_0-norm minimization into l_1-norm minimization [41]. Note that the problem in (11) aims at obtaining a sparse solution for the imaging scene. Nevertheless, the imaging scene is not always sparse in SAR applications.

In this article, three video SAR imaging results datasets are applied to analyze the property of the data. The first one is Eubank Gate and Traffic Video SAR imaging results by Sandia National Laboratories [42]. The second one is Solar Tower Video SAR results by Sandia National Laboratories [42]. The third one is the formulated video SAR data from the SAR image by Airbus Defence and Space [43]. First, we analyze the sparsity of the three video SAR data. The metric to assess the sparsity property is the fraction of total magnitude captured by the top portion elements of the sparse vector σ to the total magnitude of σ. As a result, the metric is defined as the empirical cumulative distribution function (CDF) of the values of vector σ, which is expressed as

$$\text{CDF}_s(\text{pe}) = \frac{\sum_{k=1}^{\text{pe}\cdot M\times N\times T} \sigma(k)}{\sum_{k=1}^{M\times N\times T} \sigma(k)} \tag{12}$$

where $\sigma = \text{vec}(\mathcal{X})$, $\sigma(k)$ is the magnitude of σ with the kth largest value. To solve the problem of nonsparse imaging scene, some sparse transformations, including discrete Fourier transformation (DFT) and discrete cosine transformation (DCT), which can obtain the sparse representation of imaging scene, are applied. Here, the CDFs of the sparsity with original video SAR data and under different sparse transformations are shown in the first column of Fig. 2. It can be found that top 84% values capture 90% magnitude of $\mathcal{X}$ for video SAR dataset 1, while the ratio decreases to 46% for the video SAR data under DCT.

However, the correlation information of the imaging scene is ignored by simply processing the imaging scene as a sparse or compressible vector. In [38], to exploit the correlation structure of the video SAR tensor $\mathcal{X}$, the authors applied low rankness as the feature to model $\mathcal{X}$. $\mathcal{X}$ can be recovered by solving the following rank minimization problem, namely:

$$\begin{aligned} &\arg \min_{\mathcal{X}} \ \text{rank}(\mathcal{X}) \\ &\text{s.t. } \mathcal{Y}_d(:,:,t) = \mathbf{\Theta}_a \Re(\mathcal{X}(:,:,t))\mathbf{\Theta}_r, \quad t = 1, 2, \ldots, T. \end{aligned} \tag{13}$$

The low-rankness property of the three video SAR datasets is also presented. The metric to assess the low-rankness property is the fraction of total energy captured by the top portion of singular tubes. The CDF of the singular values of tensor $\mathcal{X}$ is defined as $\text{CD}F_l(\text{pe})$, which is expressed as follows:

$$\text{CD}F_l(\text{pe}) = \frac{\sum_{k=1}^{\text{pe}\cdot\min(M,N)} \mathcal{M}(k,k,1)}{\sum_{k=1}^{\min(M,N)} \mathcal{M}(k,k,1)} \tag{14}$$

where $\mathcal{M}(k,k,1)$ is the first frontal slice of $\mathcal{M}$ with the kth largest value. Note that $\mathcal{M}$ can be calculated by tensor singular value decomposition (t-SVD), which is given in Algorithm 1 (for more details, please refer to [44]). The CDFs of the low rankness of the three video SAR datasets are shown in the second column of Fig. 2. We can find that the top 37% singular values capture 90% energy of $\mathcal{X}$ for video SAR data 1. Compared with sparse property, the video SAR data 1 can be represented with a smaller portion of tensor singular values.

In practical applications, the standard rank minimization imaging algorithms suffer performance degradation when the imaging scene exists a small portion of anomaly dominant scatterers. In these cases, the low rankness of video SAR data is broken down by the anomaly dominant scatterers.

Algorithm 1 t-SVD

Input: $\mathcal{X} \in \mathbb{C}^{M\times N\times T}$.
Output: T-SVD components $\mathcal{U}$, $\mathcal{M}$ and $\mathcal{V}$ of $\mathcal{X}$.

Compute $\bar{\mathcal{X}} = \textbf{fft}(\mathcal{X}, [\ ], 3)$.
Compute each fontal slice of $\bar{\mathcal{U}}$, $\bar{\mathcal{M}}$ and $\bar{\mathcal{V}}$ from $\bar{\mathcal{X}}$.
for $t = 1, \cdots, \lceil \frac{t+1}{2} \rceil$ **do**
  $[\bar{\mathcal{U}}^{(t)}, \bar{\mathcal{M}}^{(t)}, \bar{\mathcal{V}}^{(t)}] = SVD(\bar{\mathcal{X}}^{(t)})$
end for
for $t = \lceil \frac{t+1}{2} \rceil + 1, \cdots, T$ **do**
  $\bar{\mathcal{U}}^{(t)} = \textbf{conj}(\bar{\mathcal{U}}^{(T-t+2)})$;
  $\bar{\mathcal{M}}^{(t)} = (\bar{\mathcal{M}}^{(T-t+2)})$;
  $\bar{\mathcal{V}}^{(t)} = \textbf{conj}(\bar{\mathcal{V}}^{(T-t+2)})$;
end for
Compute $\mathcal{U} = \textbf{ifft}(\bar{\mathcal{U}}, [\ ], 3)$, $\mathcal{M} = \textbf{ifft}(\bar{\mathcal{M}}, [\ ], 3)$ and $\mathcal{V} = \textbf{ifft}(\bar{\mathcal{V}}, [\ ], 3)$.

In this article, instead of utilizing the sparsity and low-rankness of the video SAR data separately, we model the

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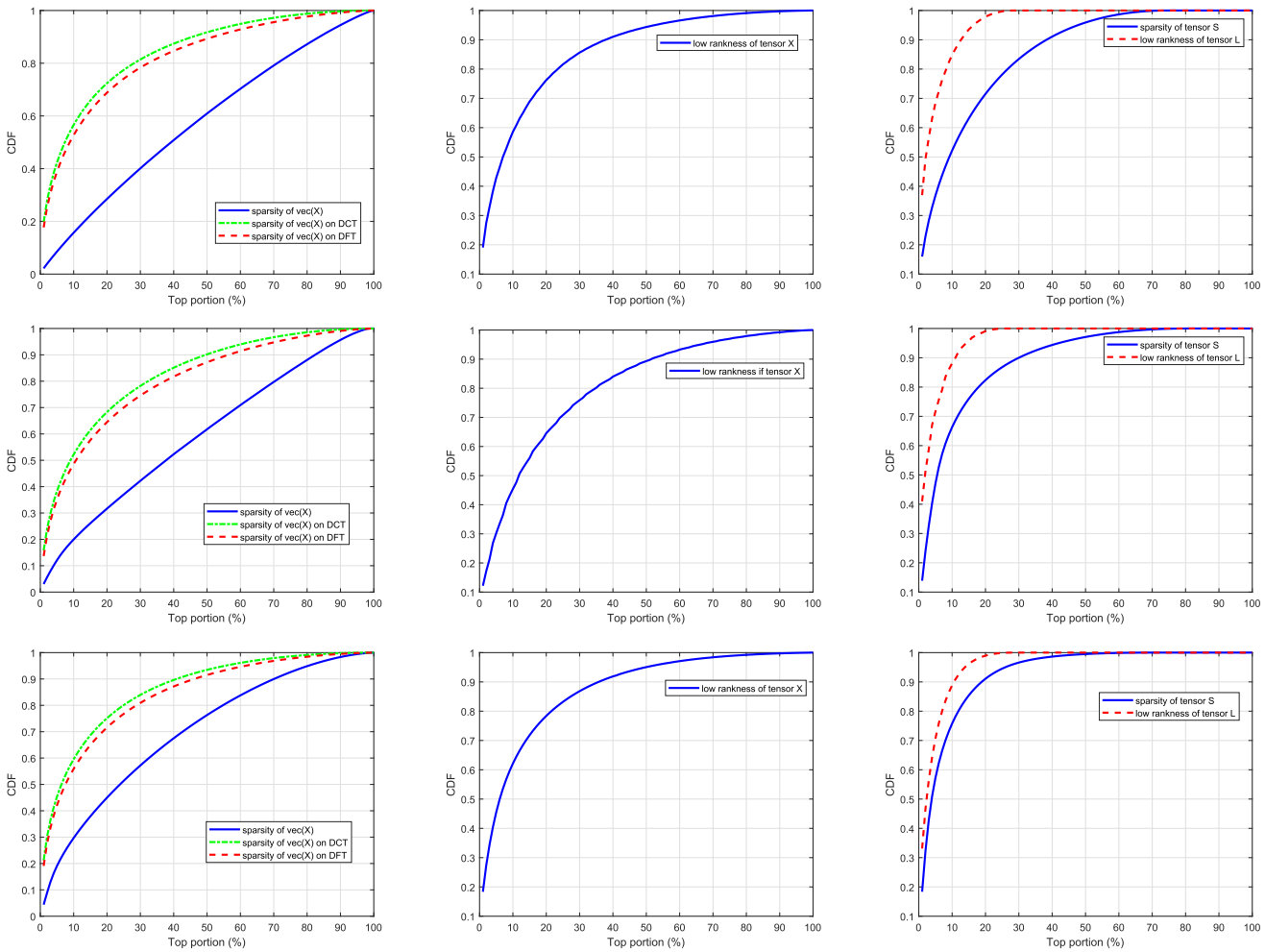


Fig. 2. Analysis of the video SAR scene property. Columns 1–3 correspond to the CDFs of the sparsity, the low rankness, and the joint sparsity and low-rankness model, respectively. Rows 1–3 correspond to the video SAR scenes 1–3, respectively.

original data $\mathcal{X}$ as the sum of the sparse tensor $\mathcal{S}$, i.e., the anomaly dominant scatterers of the scene, and the low-rank tensor $\mathcal{L}$ that represents the correlative background of the imaging scene. The above relation can be written as

$$\mathcal{X} = \mathcal{L} + \mathcal{S}. \tag{15}$$

In the above tensor decomposition model, the original video data tensor $\mathcal{X}$ is split into two parts, namely, sparse tensor $\mathcal{S}$ and low-rank tensor $\mathcal{L}$. To obtain the separated parts, an effective and efficient decomposition algorithm for tensor named tensor robust principal component analysis (TRPCA) is applied [44]. The CDFs of the low rankness of $\mathcal{L}$ and sparsity of $\mathcal{S}$ for the three video SAR datasets are shown in the third column of Fig. 2. It can be observed that the top 12% singular values capture 90% energy of $\mathcal{L}$ video SAR data 1, which means that a smaller portion of singular values can be applied to represent the $\mathcal{L}$. As for the CDF of the sparsity of $\mathcal{S}$, top 38% values capture 90% magnitude of $\mathcal{S}$. Although the *pe* percent to capture the almost magnitude of the sparse part is not decreased significantly, it should be noted that the total magnitude of the separated sparse tensor $\mathcal{S}$ captures 4.5% total magnitude of the original tensor $\mathcal{X}$. The total magnitude of the separated sparse tensor $\mathcal{S}$ captures 13.72% total magnitude of the original tensor $\mathcal{X}$ for video SAR data 2. As for video SAR data 3, the total magnitude of the separated sparse tensor captures 13.87% total magnitude of the original tensor. It can be found that only a very small portion of the original data is neglected with the joint sparse and low-rank model.

From the above analysis, we can find that the joint sparse and low-rank tensor model is more efficient for the video SAR data compared with the separate sparse tensor or low-rank tensor model. To fully exploit the property of the video SAR data $\mathcal{X}$, the video SAR imaging problem is modeled as the following joint rank and l_0-norm minimization problem:

$$\begin{aligned} &\arg\min_{\mathcal{L},\mathcal{S}} \ \operatorname{rank}(\mathcal{L}) + \lambda\|\mathcal{S}\|_0 \\ &\text{s.t. } \mathcal{Y}_d(:,:,t) = \boldsymbol{\Theta}_a \Re(\mathcal{L}(:,:,t) + \mathcal{S}(:,:,t))\boldsymbol{\Theta}_r \\ &\quad t = 1, 2, \ldots, T. \end{aligned} \tag{16}$$

III. Joint Low-Rank and Sparse Tensors Recovery Method

In Section II, the observation model for video SAR imaging is first established. After that, the video SAR imaging problem

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is modeled as a joint low-rank and sparse tensors recovery problem based on the property analysis of the real video SAR data. Although there are some efficient tensor analysis methods for joint low-rank and sparse tensors [44]–[47], most of them focus on the decomposition problem or TRPCA rather than the recovery problem of tensors from reduced measurements. In this section, a tensor recovery method based on ADMM for video SAR imaging is proposed where the efficient observation model can be applied.

Since the joint recovery problem in (16) is an NP-hard problem, the l_0-norm minimization problem and rank minimization problem in (16) is first converted into the l_1-norm minimization problem and the tensor nuclear norm minimization problem, respectively, which is given by

$$\begin{aligned} &\arg\min_{\mathcal{L},\mathcal{S}} \ \|\mathcal{L}\|_* + \lambda\|\mathcal{S}\|_1 \\ &\text{s.t. } \mathcal{Y}_d(:,:,t) = \mathbf{\Theta}_a\mathfrak{R}(\mathcal{L}(:,:,t) + \mathcal{S}(:,:,t))\mathbf{\Theta}_r \\ &\quad t = 1, 2, \ldots, T \end{aligned} \tag{17}$$

where $\|\cdot\|_*$ represents the tensor nuclear norm, which is the sum of tensor singular values [44].

The augmented Lagrangian associated with (17) is given as follows:

$$\begin{aligned} &L(\mathcal{L}, \mathcal{S}, \mathcal{Q}, \rho) \\ &= \|\mathcal{L}\|_* + \lambda\|\mathcal{S}\|_1 \\ &\quad + \langle \mathcal{Q}, \Theta_a\mathfrak{R}(\mathcal{L}+\mathcal{S})\Theta_r - \mathcal{Y}_d\rangle + \frac{\rho}{2}\|\Theta_a\mathfrak{R}(\mathcal{L}+\mathcal{S})\Theta_r - \mathcal{Y}_d\|_F^2 \end{aligned} \tag{18}$$

where $\mathcal{Q}$ is the Lagrange multipliers, $\langle\cdot,\cdot\rangle$ is the tensor inner product, and ρ is the penalty parameter. Note that the formation of $\mathfrak{R}(\mathcal{L}+\mathcal{S})$ is achieved by calculating each frontal slice separately.

In the ADMM method, we separate the two variables into two blocks: block 1 contains $\mathcal{L}$ and block 2 contains $\mathcal{S}$. In this way, we can update blocks 1 and 2 alternatively by fixing another in the Lagrangian.

1) Update of $\mathcal{L}$: First, the updating scheme of low-rank tensor $\mathcal{L}$ is achieved by

$$\mathcal{L}_{g+1} = \arg\min_{\mathcal{L}} \|\mathcal{L}\|_* + \frac{\rho_g}{2}\|\mathcal{L} - \mathcal{L}_g + \mathcal{A}_g\|_F^2 \tag{19}$$

where the calculation of intermediate variable $\mathcal{A}_g$ is given in (20), as shown at the bottom of the page. In (20), $\eta_{\mathfrak{R}} = \|\Theta_a\mathfrak{R}(\mathbf{I})\Theta_r\|_2^2$. Note that the tth frontal slice of $\mathcal{A}_g$, i.e., $\mathcal{A}_g^{(t)}$, is formed by the imaging processing and the inverse processing of the chirp scaling algorithm, which are denoted as $\mathfrak{I}(\cdot)$ and $\mathfrak{R}(\cdot)$, respectively.

Note that the update of $\mathcal{L}$ requires computing the proximal operator of tensor nuclear norm, i.e., tensor singular value thresholding (t-SVT). Let $\mathcal{X} = \mathcal{U} * \mathcal{M} * \mathcal{V}^*$ be the t-SVD of $\mathcal{X}$. The t-SVT operator is given as follows:

$$\mathcal{D}_\delta(\mathcal{X}) = \mathcal{U} * \mathcal{M}_\delta * \mathcal{V}^* \tag{21}$$

where

$$\mathcal{M}_\delta = \mathbf{ifft}\Big(\big(\bar{\mathcal{M}} - \delta\big)_+, [\,], 3\Big). \tag{22}$$

Note that x_+ denotes the positive part of x, i.e., $x_+ = \max(x, 0)$.

2) Update of $\mathcal{S}$: The updating scheme of sparse tensor $\mathcal{S}$ is achieved by

$$\mathcal{S}_{g+1} = \arg\min_{\mathcal{S}} \lambda\,\|\mathcal{S}\|_1 + \frac{\rho_g}{2}\|\mathcal{S} - \mathcal{S}_g + \mathcal{B}_g\|_F^2 \tag{23}$$

where the calculation of intermediate variable $\mathcal{B}_g$ is given in (26), as shown at the bottom of the page. Similarly, the tth frontal slice of $\mathcal{B}_g$, i.e., $\mathcal{B}_g^{(t)}$, is also formed by the imaging processing and the inverse processing of the chirp scaling algorithm. Note that the value of the updated sparse tensor can be obtained by applying soft-thresholding operator, which is defined as follows:

$$\eta_\delta(\mathcal{X}) = \text{sign}(\mathcal{X})(|\mathcal{X}| - \delta)_+ \tag{24}$$

where

$$sign(x) = \begin{cases} x/|x|, & \text{for } x \neq 0 \\ 0, & \text{for } x = 0. \end{cases} \tag{25}$$

3) Update Lagrange Multiplier $\mathcal{Q}$: The update of the Lagrange multiplier is achieved for each frontal slice (each frame) separately. For the tth frontal slice of $\mathcal{Q}$, the update scheme is given as follows:

$$\mathcal{Q}_{g+1}^{(t)} = \mathcal{Q}_g^{(t)} + \rho_g\Big(\Theta_a\mathfrak{R}\Big(\mathcal{L}_{g+1}^{(t)} + \mathcal{S}_{g+1}^{(t)}\Big)\Theta_r - \mathcal{Y}_d^{(t)}\Big). \tag{27}$$

After obtaining all the frontal slices, the Lagrange multiplier $\mathcal{Q}$ can be formulated.

4) Update Penalty Parameter ρ: The adaptive updating strategy for the penalty parameter is

$$\rho_{g+1} = \min\big(\alpha\rho_g, \rho_{\max}\big) \tag{28}$$

where $\rho_{\max}$ is an upper bound of ρ and $\alpha \geq 1$ is a constant.

If the update rate of both the low-rank and sparse tensors is less than a predefined value, the iteration is stopped. Finally, the recovery of the video SAR data is achieved. From the iteration steps of the ADMM method, it can be found that there only includes processing and inverse processing steps of the chirp scaling algorithm, where the storage of the observation model tensor in [38], which requires huge storage capacity, can be avoided.

$$\mathcal{A}_g^{(t)} = \mathfrak{I}\left(\Theta_a^T\left(\Theta_a\mathfrak{R}\big(\mathcal{L}_g^{(t)} + \mathcal{S}_g^{(t)}\big)\Theta_r - \mathcal{Y}_d^{(t)} + \frac{\mathcal{Q}_g^{(t)}}{\rho_g}\right)\Theta_r^T\right)\Big/\eta_{\mathfrak{R}}. \tag{20}$$

$$\mathcal{B}_g^{(t)} = \mathfrak{I}\left(\Theta_a^T\left(\Theta_a\mathfrak{R}\Big(\mathcal{L}_{g+1}^{(t)} + \mathcal{S}_g^{(t)}\Big)\Theta_r - \mathcal{Y}_d^{(t)} + \frac{\mathcal{Q}_g^{(t)}}{\rho_g}\right)\Theta_r^T\right)\Big/\eta_{\mathfrak{R}}. \tag{26}$$

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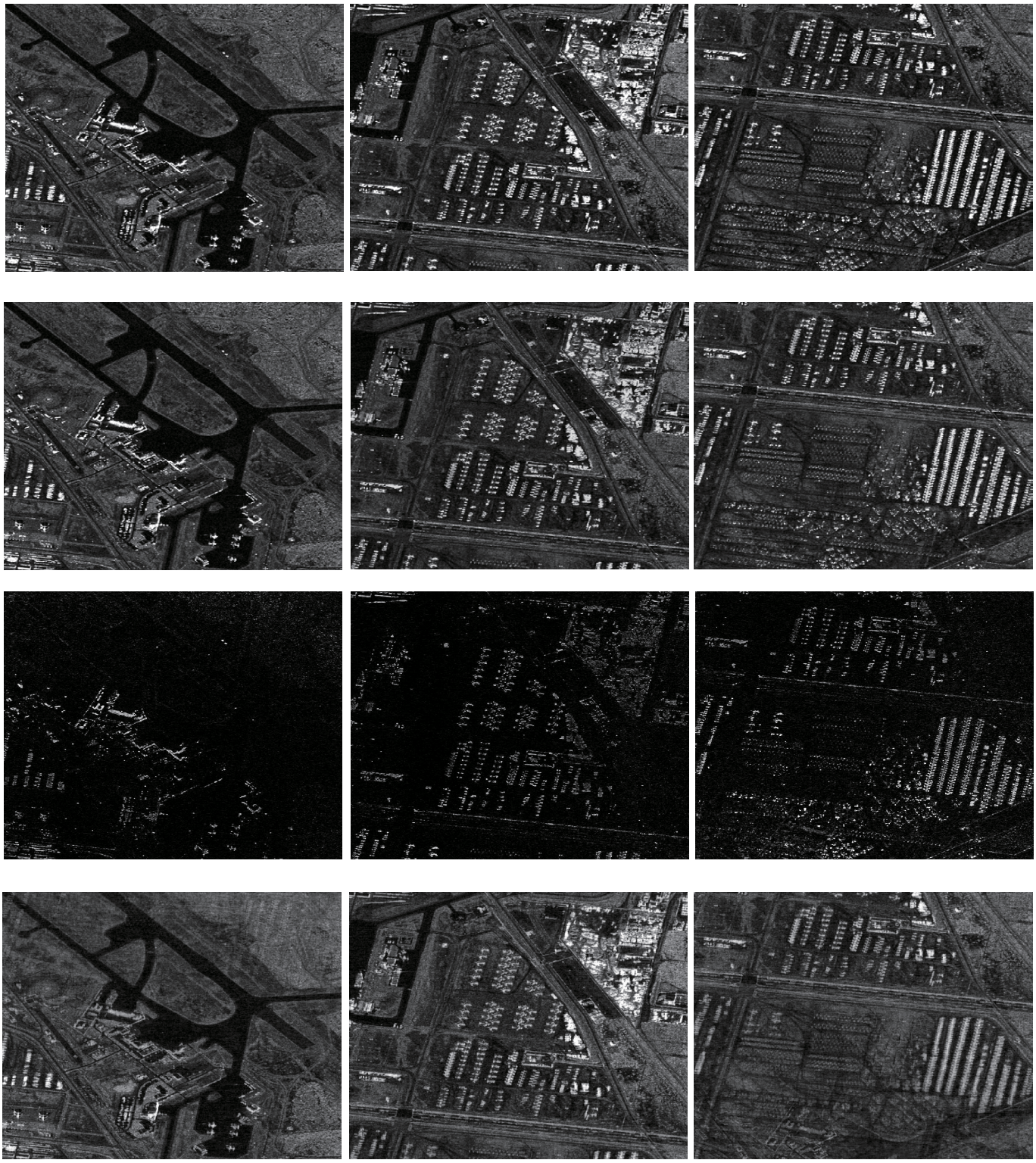


Fig. 3. Imaging results obtained by the proposed method. Columns 1–3 correspond to the 5th, 22nd, and 30th frames, respectively. Rows 1–4 correspond to the original results, the proposed method, the sparse part obtained by the proposed method, and the low-rank part obtained by the proposed method, respectively.

IV. Numerical Results and Discussion

In this section, numerical results are presented to validate the effectiveness of the proposed video SAR imaging method. The state-of-the-art video SAR imaging method based on low-rank tensor recovery [38] is implemented for comparisons. Moreover, we compare the performance of the proposed method with matrix recovery methods, including the sparse matrix recovery method [48], the low-rank matrix recovery method, and the joint sparse and low-rank matrices recovery method [34].

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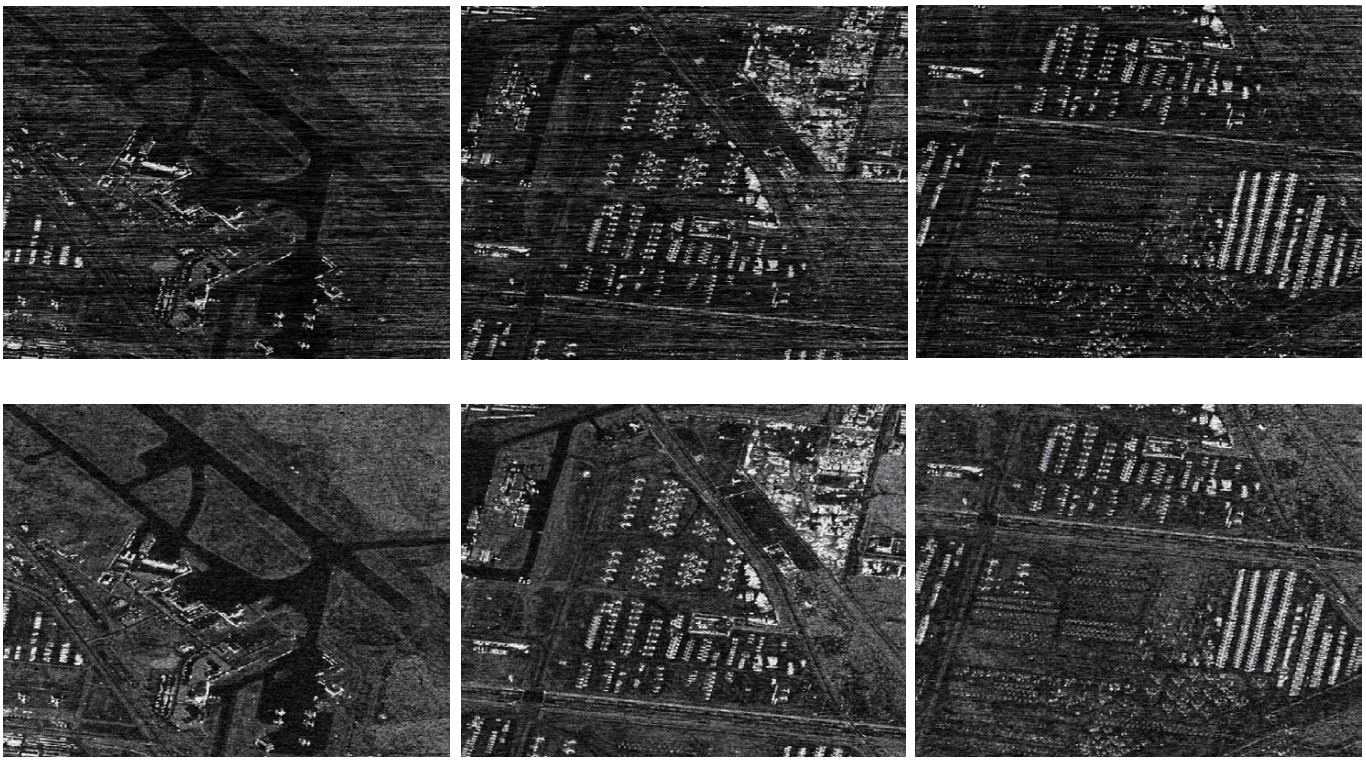


Fig. 4. Imaging results obtained by different methods. Columns 1–3 correspond to the 5th, 22nd, and 30th frames, respectively. Rows 1 and 2 correspond to sparse tensor recovery and low-rank tensor recovery methods, respectively.

TABLE I
SIMULATION PARAMETERS FOR VIDEO SAR

Parameter	Value
Carrier frequency	10 GHz
Signal bandwidth	400 MHz
Pulse width	1 us
Fast time sampling rate	500 MHz
Synthetic aperture time	2 s
Radar platform velocity	200 m/s
Radar position in x direction at azimuth central time	-3 km
Radar position in y direction at azimuth central time	0 km
Radar position in z direction at azimuth central time	5 km

A. Tensor Recovery for Video SAR Scene 1

First, simulations for video scene 1 are conducted to verify the effectiveness of the proposed method where the detailed simulation parameters are listed in Table I. The original SAR video is formed by the SAR image and the SAR video has 32 frames. The consecutive frames have 93.75% overlap. The sampling rate is 60 % in azimuth, that is, only 60% samples or data are applied to recover the original SAR video. The values of $\rho_{\max}$ and α in the tensor ADMM method are 10^8 and 1.1, respectively.

The selected frames of the recovered SAR video by the proposed method are shown in the second row of Fig. 3. To clearly illustrate the recovery performance of the proposed method, the corresponding original video SAR frames are shown in the first row of Fig. 3. It can be found that the proposed method can achieve a good reconstruction of the original SAR video. Moreover, the sparse part and the low-rank part of selected frames obtained by the proposed method in video SAR scene 1 simulation are shown in the third and fourth rows of Fig. 3, respectively. It can be found that the sparse part can represent the dominant scatterers in the SAR video correctly and the low-rank part accords with the background of the SAR video.

The state-of-the-art video SAR imaging method based on low-rank tensor recovery proposed in [38] is also applied to reconstruct the original SAR video for comparison. The corresponding frames obtained by the low-rank tensor recovery method are shown in the second row of Fig. 4. Compared with the results obtained by the proposed method, the recovered SAR frames are a bit blurry, especially the strong scatterers. This is because this method assumes that the SAR video is low-rank. However, SAR video with strong scatterers is generally not low-rank when the observed scene varies with respect to the observed time. Therefore, the low-rank hypothesis destroys the original scattering characteristics of strong scatterers. Besides, the corresponding frames of the SAR video recovered by the tensor recovery method only utilizing the sparse property are shown in the first row of Fig. 4. Compared with the results obtained by the proposed method and the low-rank tensor recovery method, the method based on sparse tensor recovery is more inclined to recover the strong scatterers in the SAR video and suppress the low-rank background part at the same time.

To quantitatively compare the reconstruction performance of different tensor recovery methods, we introduce a peak signal-to-noise ratio (PSNR) to analyze the reconstruction quality of the tth frame of the video SAR, which can be calculated by

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{\max(|\mathcal{X}(:,:,t)_{\mathrm{or}}|)}{\mathrm{RMSE}}\right). \tag{29}$$

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TABLE II
IMAGING QUALITY FOR VIDEO SCENE 1 SIMULATIONS WITH DIFFERENT TENSOR RECOVERY METHODS

Frames	1st	2nd	3rd	4th	5th	6th	7th	8th
Sparse tensor recovery method	14.72 dB	14.98 dB	15.06 dB	15.21 dB	15.32 dB	15.37 dB	15.35 dB	15.33 dB
Low-rank tensor recovery method	19.38 dB	20.19 dB	20.99 dB	21.55 dB	22.01 dB	22.21 dB	22.29 dB	22.44 dB
Proposed recovery method	26.13 dB	27.90 dB	29.30 dB	30.39 dB	31.26 dB	31.70 dB	32.03 dB	32.28 dB
Frames	9th	10th	11th	12th	13th	14th	15th	16th
Sparse tensor recovery method	15.27 dB	15.24 dB	15.12 dB	14.96 dB	14.78 dB	14.60 dB	14.48 dB	14.30 dB
Low-rank tensor recovery method	22.38 dB	22.40 dB	22.35 dB	22.31 dB	22.30 dB	22.28 dB	22.30 dB	22.26 dB
Proposed recovery method	32.32 dB	32.38 dB	32.22 dB	32.15 dB	32.00 dB	31.85 dB	31.66 dB	31.66 dB
Frames	17th	18th	19th	20th	21st	22nd	23rd	24th
Sparse tensor recovery method	14.30 dB	14.22 dB	14.28 dB	14.42 dB	14.45 dB	14.47 dB	14.48 dB	14.45 dB
Low-rank tensor recovery method	22.19 dB	22.10 dB	22.02 dB	22.02 dB	21.86 dB	21.72 dB	21.67 dB	21.48 dB
Proposed recovery method	31.53 dB	31.30 dB	31.31 dB	31.18 dB	31.02 dB	30.96 dB	30.71 dB	30.36 dB
Frames	25th	26th	27th	28th	29th	30th	31st	32nd
Sparse tensor recovery method	14.59 dB	14.64 dB	14.79 dB	15.03 dB	15.18 dB	15.32 dB	15.41 dB	15.53 dB
Low-rank tensor recovery method	21.23 dB	21.03 dB	20.72 dB	20.42 dB	20.15 dB	19.76 dB	19.22 dB	18.95 dB
Proposed recovery method	29.98 dB	29.56 dB	29.10 dB	28.72 dB	28.15 dB	27.44 dB	26.39 dB	25.16 dB

TABLE III
IMAGING QUALITY FOR VIDEO SCENE 2 SIMULATIONS WITH DIFFERENT TENSOR RECOVERY METHODS

Frames	1st	2nd	3rd	4th	5th	6th	7th	8th
Sparse tensor recovery method	26.47 dB	26.40 dB	26.07 dB	25.53 dB	25.21 dB	24.60 dB	24.78 dB	24.87 dB
Low-rank tensor recovery method	24.98 dB	24.88 dB	24.85 dB	24.91 dB	24.89 dB	25.03 dB	25.45 dB	25.87 dB
Proposed recovery method	32.35 dB	32.84 dB	33.01 dB	32.87 dB	32.77 dB	32.26 dB	32.61 dB	32.80 dB
Frames	9th	10th	11th	12th	13th	14th	15th	16th
Sparse tensor recovery method	25.12 dB	25.39 dB	25.51 dB	25.48 dB	25.64 dB	26.27 dB	26.53 dB	27.26 dB
Low-rank tensor recovery method	26.25 dB	26.59 dB	27.06 dB	27.46 dB	28.05 dB	28.62 dB	29.63 dB	30.27 dB
Proposed recovery method	33.12 dB	33.50 dB	33.88 dB	34.14 dB	34.65 dB	35.51 dB	36.71 dB	38.64 dB
Frames	17th	18th	19th	20th	21st	22nd	23rd	24th
Sparse tensor recovery method	25.19 dB	26.95 dB	26.72 dB	26.40 dB	26.39 dB	26.44 dB	26.26 dB	26.14 dB
Low-rank tensor recovery method	28.64 dB	30.76 dB	30.81 dB	30.79 dB	30.85 dB	30.80 dB	30.64 dB	30.48 dB
Proposed recovery method	36.71 dB	38.47 dB	38.16 dB	37.89 dB	37.93 dB	37.97 dB	37.79 dB	37.48 dB
Frames	25th	26th	27th	28th	29th	30th	31st	32nd
Sparse tensor recovery method	26.09 dB	25.99 dB	25.93 dB	25.84 dB	25.78 dB	25.69 dB	25.58 dB	25.73 dB
Low-rank tensor recovery method	30.23 dB	30.04 dB	29.61 dB	29.17 dB	28.62 dB	27.97 dB	27.17 dB	26.32 dB
Proposed recovery method	37.22 dB	36.92 dB	36.48 dB	35.92 dB	35.38 dB	34.58 dB	33.69 dB	33.08 dB

Note that RMSE is the root-mean-square error of the tth recovered frame and can be calculated by

$$\mathrm{RMSE} = \sqrt{\frac{1}{MN}\|\mathcal{X}(:,:,t)_{\mathrm{or}} - \mathcal{X}(:,:,t)_{\mathrm{re}}\|_F^2} \tag{30}$$

where $\mathcal{X}(:,:,t)_{\mathrm{or}}$ denotes the original reflectivity coefficients of the tth frame of the SAR video and $\mathcal{X}(:,:,t)_{\mathrm{re}}$ represents the recovered reflectivity coefficients of the tth frame of the SAR video.

The PSNR values of all 32 frames for video scene 1 obtained by different methods are listed in Table II. It can be found that the proposed method can achieve a better recovery performance compared with the methods only separately considering the sparse or low-rank property of the SAR video. Compared with the sparse tensor recovery method, the proposed method improves the average PSNR value of multiple frames by 15.45 dB. Moreover, the proposed method can enhance the average PSNR value by 8.87 dB compared with the state-of-the-art low-rank tensor recovery method.

The PSNR performances of the proposed method, the low-rank tensor recovery method, and the sparse tensor recovery method are shown in Fig. 5 as a function of the subsampling rate where 20%–90% sampling rates are applied. It can be found that the recovered performance degrades when the subsampling rate decreases. Note that the proposed method can achieve a better imaging performance compared with the sparse tensor recovery and low-rank tensor recovery methods for different subsampling rates.

Generally speaking, the imaging processing time of the proposed method is closely related to the dimension of the video SAR echo, the imaging processing method, the number of iterations, and the processing platform used. The computational complexity of the decoupling model for each frame in the proposed method is consistent with the chirp scaling imaging method. In this article, the processing steps of chirp scaling method and the inverse process of chirp scaling are required multiple times due to the iterative operations and the multiple frames of SAR video. To accelerate the speed of processing, the range compression function, azimuth compression function, and chirp scaling function required for

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TABLE IV
IMAGING QUALITY WITH DIFFERENT TENSOR RECOVERY AND MATRIX RECOVERY METHODS

Frames	1st	2nd	3rd	4th	5th	6th	7th	8th
CS-based method	14.72 dB	14.97 dB	15.21 dB	15.25 dB	15.13 dB	15.01 dB	14.73 dB	14.43 dB
MS-based method	21.71 dB	21.46 dB	20.97 dB	20.29 dB	19.79 dB	18.90 dB	18.73 dB	18.60 dB
Joint sparse and low-rank matrices method	23.68 dB	23.92 dB	23.81 dB	23.23 dB	22.76 dB	22.29 dB	21.61 dB	21.58 dB
Sparse tensor recovery method	14.78 dB	15.11 dB	15.26 dB	15.27 dB	15.26 dB	15.05 dB	14.86 dB	14.48 dB
Low-rank tensor recovery method	18.23 dB	18.81 dB	19.15 dB	19.14 dB	19.20 dB	18.93 dB	18.94 dB	18.81 dB
Proposed recovery method	25.74 dB	27.33 dB	28.00 dB	28.13 dB	28.10 dB	27.71 dB	27.35 dB	27.03 dB
Frames	9th	10th	11th	12th	13th	14th	15th	16th
CS-based method	14.17 dB	14.27 dB	14.29 dB	14.41 dB	14.58 dB	14.70 dB	15.21 dB	15.41 dB
MS-based method	18.25 dB	18.22 dB	18.20 dB	18.28 dB	18.30 dB	18.34 dB	18.84 dB	19.09 dB
Joint sparse and low-rank matrices method	21.35 dB	21.24 dB	21.23 dB	21.39 dB	21.48 dB	21.55 dB	22.24 dB	22.65 dB
Sparse tensor recovery method	14.24 dB	14.32 dB	14.33 dB	14.55 dB	14.58 dB	14.83 dB	15.17 dB	15.52 dB
Low-rank tensor recovery method	18.73 dB	18.58 dB	18.51 dB	18.38 dB	18.11 dB	17.88 dB	17.68 dB	17.55 dB
Proposed recovery method	26.87 dB	26.56 dB	26.48 dB	26.40 dB	25.94 dB	25.48 dB	25.17 dB	24.35 dB

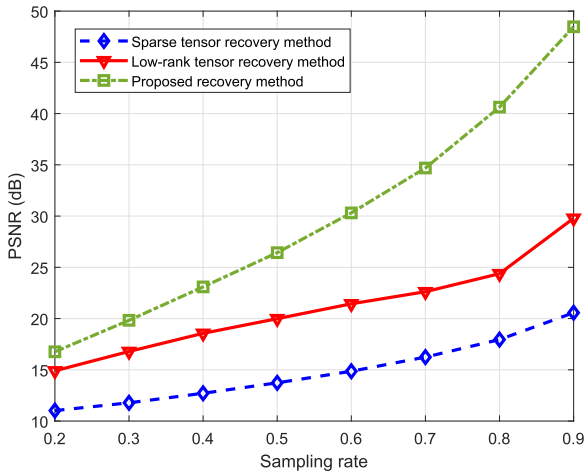


Fig. 5. Achievable PSNR values of the different methods with respect to subsampling rate.

the chirp scaling imaging algorithm can be calculated and stored in advance. As for the computational time required to obtain the final SAR video in video SAR scene 1 simulation, the computing time of the proposed method is 466.76 s on a core 3.30-GHz Inter PC. Note that the calculations of intermediate variables $\mathcal{A}_g$ and $\mathcal{B}_g$, as shown in (20) and (26), for different frames in the same iteration are independent of each other. Thus, the imaging of multiple frames in the video SAR can be processed in a parallel processing platform, such as GPU, which can greatly reduce the processing time in practical applications.

B. Tensor Recovery for Video SAR Scene 2

To further illustrate the effectiveness of the proposed method, we use another video SAR data as the input for simulation. The original SAR video is 32 frames with 90% overlap between the frames. The simulation parameters are the same as the previous simulation. The undersampling rate of azimuth is 35%, which means that the total data volume of the subsampled echo is only 35% of the original data. Fig. 6 shows the 1st, 10th, and 26th frame results obtained by the sparse tensor recovery method, the low-rank tensor recovery method, and the proposed method, respectively. Besides, the corresponding frames of the original SAR video are shown in the first row of Fig. 6. It can be seen that the imaging results obtained by the proposed method are better than the low-rank tensor and sparse tensor recovery methods. Meanwhile, the PSNR values of reconstructed SAR video obtained by different methods are shown in Table III. It can be seen that the proposed method can achieve 9.35- and 7.18-dB average PSNR values improvement compared with the sparse tensor and low-rank tensor recovery methods, respectively, which shows the effectiveness of the proposed method.

C. Comparisons With Frame-by-Frame Recovery Methods

We compare the reconstruction performance of tensor recovery methods with the matrix recovery imaging methods. The original SAR video is 16 frames with 87.5% overlap between frames. The simulation parameters are the same as simulation 1, which are given in Table I. The undersampling rate in azimuth is 60%.

The matrix recovery method obtains the SAR video by reconstructing each frame separately and then synthesizes the reconstructed multiple frames into the final SAR video. Table IV shows the PSNR values of the multiple frames obtained by the matrix recovery methods, i.e., the sparse matrix [48], low-rank matrix, the state-of-the-art joint sparse and low-rank matrices recovery method [34], and the tensor recovery methods, including the sparse tensor, low-rank tensor, and the proposed method. Compared with the results obtained by the CS method, i.e., sparse matrix recovery method, the reconstructed performance of the sparse tensor recovery method has limited improvement. This is because the goal of sparse recovery methods is to reconstruct the dominant targets. With this kind of method, the performance will not be greatly improved through multiframe joint processing. On the other hand, we can find that the performance of the low-rank tensor method is not superior to the low-rank matrix method (MS-based method). The performance of some frames obtained by the low-rank tensor method is even worse than that of the low-rank matrix method, for example, the 1st and 15th frames. This is because the dominant scatterers in the SAR video destroy its correlation property and the distribution of the dominant scatterers varies with frame in strip mode. For the low-rank matrix method, it devotes to find the best imaging results for each frame separately. As for the

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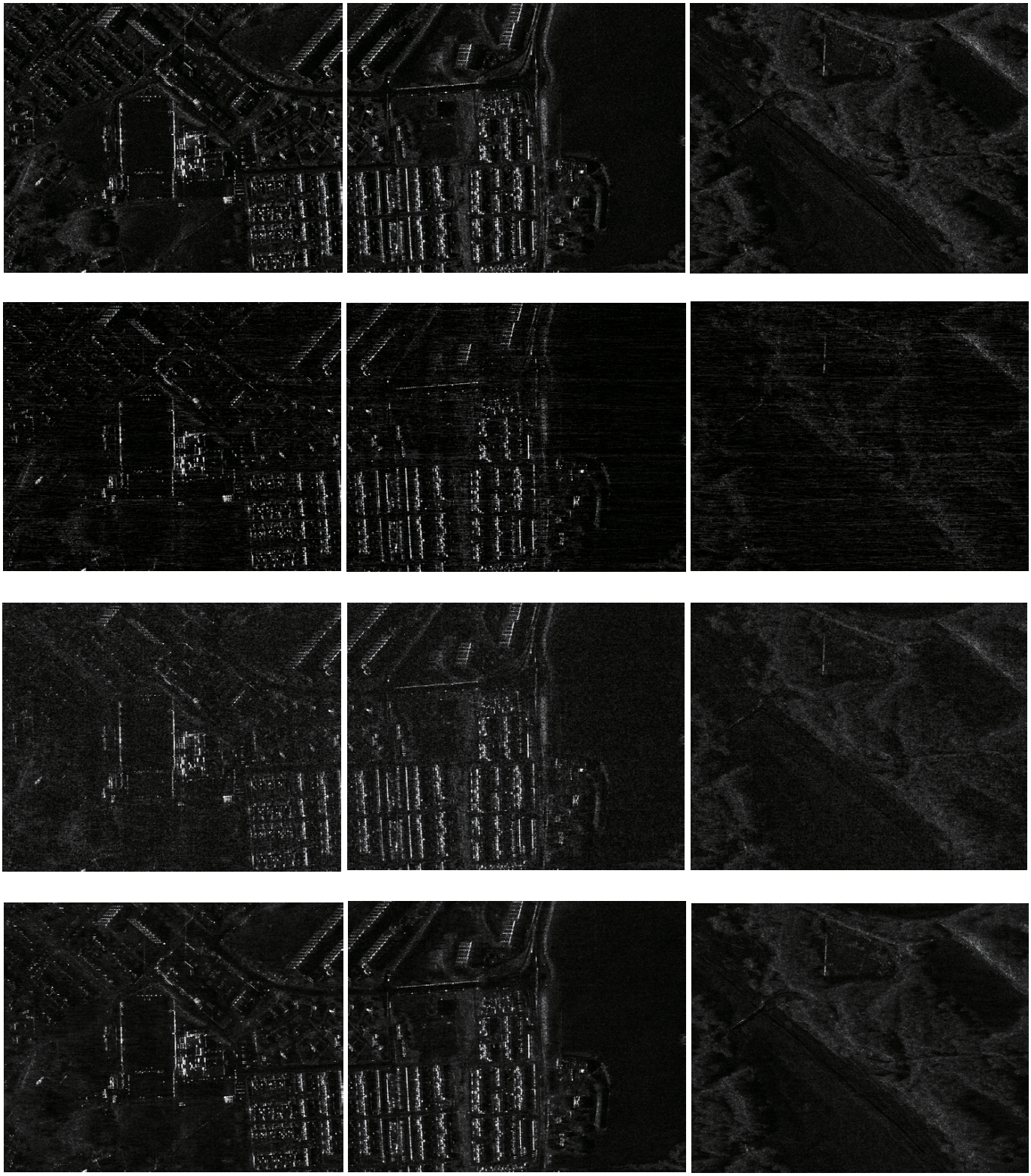


Fig. 6. Imaging results obtained by different methods. Columns 1–3 correspond to the 1st, 10th, and 26th frames, respectively. Rows 1–4 correspond to the original results, sparse tensor, low-rank tensor, and proposed methods, respectively.

low-rank tensor recovery method, the aim of this method is to obtain the best imaging results for the whole SAR video. Therefore, the change of the dominant scatterers distributions in different frames has a significant influence on the low-rank tensor recovery method, which results in performance degradation.

From Table IV, it can be found that the proposed method can obtain a better reconstruction performance compared with the joint sparse and low-rank matrices method. The average PSNR value of the SAR video is improved by 4.41 dB using the proposed method. The matrix recovery method reconstructs each frame separately, so the correlation characteristics

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between the frames are ignored. In the proposed method, both the sparse and low-rank properties of the SAR video are applied. Therefore, the disturbance of the dominant scatterers can be avoided and the correlation of the SAR video can be fully utilized, which greatly improves the reconstruction performance.

V. Conclusion

In this article, a joint sparse and low-rank tensors recovery method for video SAR imaging has been proposed. First, we find that, compared with using sparse or low-rank feature separately, the use of joint sparse and low-rank features can better characterize real SAR video. Therefore, the video SAR imaging problem is modeled as a joint sparse and low-rank tensor recovery problem in this article. After that, a tensor ADMM is proposed to efficiently solve the joint minimization problem. Compared with the sparse tensor recovery method, the proposed method can improve the average PSNR value of multiple frames by 15.45 dB. Moreover, the proposed method can enhance the average PSNR value by 8.87 dB compared with the state-of-the-art low-rank tensor recovery method. Compared with the state-of-the-art joint sparse and low-rank matrices recovery method, the proposed method can achieve a 4.41-dB improvement of the average PSNR value by utilizing the correlation property between the frames.

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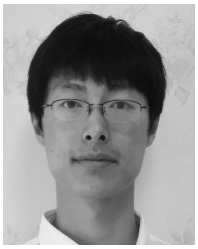
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